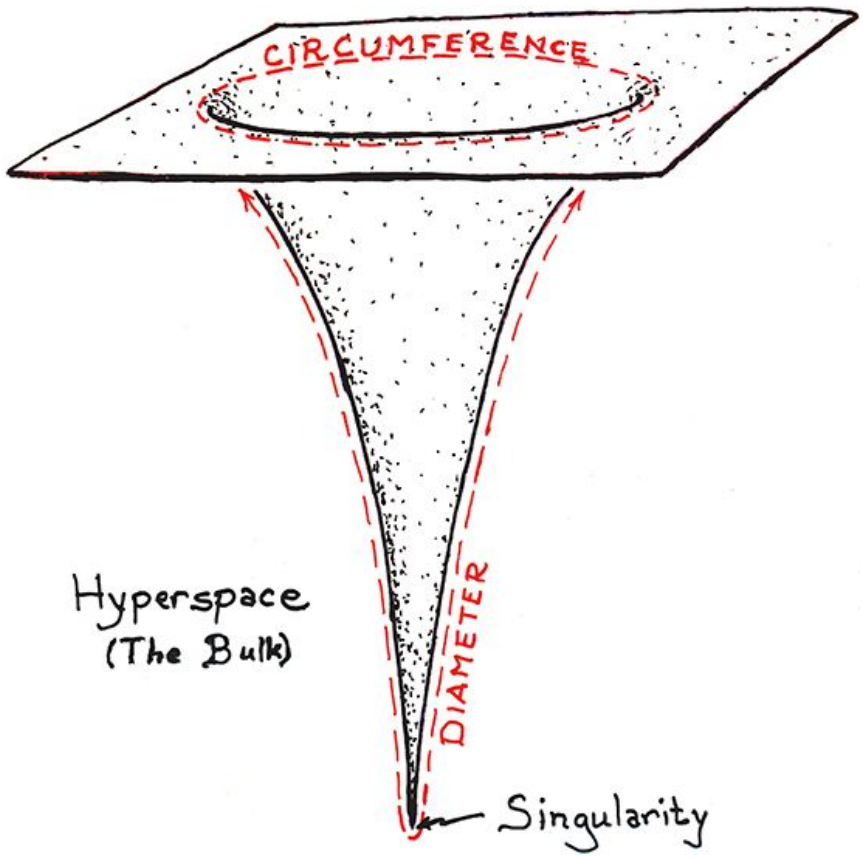


Fig. 5.2. The warped space inside and around a black hole, as seen from the bulk. *[My own hand sketch.]*

The singularity is a tiny region where the surface forms a point and thus is “infinitely warped,” and where, it turns out, tidal gravitational forces are infinitely strong, so matter as we know it gets stretched and squeezed out of existence. In chapters 26, 28, and 29, we see that Gargantua’s singularity is somewhat different from this one, and why.

For the trampoline, the warping of space is produced by the rock’s weight. Similarly, one might suspect, the black hole’s space warp is produced by the singularity at its center. Not so. In fact, the hole’s space is warped by the enormous energy of its warping. Yes, that’s what I meant to say. If this seems a bit circular to you, well, it is, but it has deep meaning.

Just as it requires a lot of energy to bend a stiff bow in preparation for shooting an arrow, so it requires a lot of energy to bend space; to